Geodesic convex optimization, first-order

Final project presentation
CSCI-GA.2945/MATH-GA.2012 Convex and Nonsmooth Optimization
Professor Michael Overton

Charles Zhu

New York University

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Introduction: generalizations of spaces

- *geodesic*: $\gamma : [0,1] \to X$ path s.t. γ parameterized by arc length, (X, d) is a metric space
- *geodesic triangle*: Δpqr, $p, q, r \in X$, formed by geodesics $\overline{pq}, \overline{qr}, \overline{rp}$, c.f. *comparison triangle*, $\Delta\bar{p}\bar{q}\bar{r}$ in k-plane (Gaussian curvature k) with same side lengths in 2D space
- *manifold* $(\mathcal{M})$: topological space s.t. each pt. is homeomorphic to pt. in Euclidean space
- *tangent space* $(T_x\mathcal{M})$: set of all tangent vectors at x
- *Riemannian manifold* $(\mathcal{M}, \mathfrak{g})$: real smooth manifold, inner product $\mathfrak{g}$ on tangent space $T_x\mathcal{M}$ s.t. for u, v vector fields on $\mathcal{M}$, $x \mapsto \langle u, v\rangle_x := \mathfrak{g}(u, v)$ is smooth function (infinitely differentiable i.e. C^∞)

Introduction: generalizations of convexity

- *geodesically convex set* (g-convexity): $f : \mathcal{M} \to \mathbb{R}$ s.t. $\forall x, y \in \mathcal{X}$, $\mathcal{X} \subset \mathcal{M}$, minimal distance geodesic γ btwn. x, y is in $\mathcal{X}$
- *geodesic strong convexity*: $f : \mathcal{M} \to \mathbb{R}$, if $\forall x, y \in \mathcal{M}$, $f(y) \geq f(x) + \langle g_x, \mathrm{Exp}_x^{-1}(y) \rangle_x + \frac{\mu}{2} d^2(x, y)$
- *geodesically L_f Lipschitz*: $\forall x, y \in \mathcal{M}$, $|f(x) - f(y)| \leq L_f d(x, y)$
- *geodesically L_g smooth*: $\forall x, y \in \mathcal{M}$, $\|g_x - \Gamma_y^x g_x\| \leq L_g d(x, y)$, where g is gradient, $\Gamma_y^x : T_y\mathcal{M} \to T_x\mathcal{M}$ is parallel transport from y to x. Furthermore $f(y) \leq f(x) + \langle g_x, \mathrm{Exp}_x^{-1}(y) \rangle_x + \frac{L_g}{2} d^2(x, y)$
- *parallel transport*: for $v \in T_X\mathcal{M}$, $\Gamma(\gamma)_y^x v$ is still a tangent vector of γ, transported to y along γ, with property $\langle u, v \rangle_x = \langle \Gamma(\gamma)_y^x u, \Gamma(\gamma)_y^x v \rangle_y$
- *exponential map* ($\mathrm{Exp}_x(v)$): mapping from $T_x\mathcal{M} \to \mathcal{M}$ s.t. $v \in T_x\mathcal{M}$ mapped to $y := \mathrm{Exp}_x(v) \in \mathcal{M}$, s.t. $\exists \gamma : [0, 1] \to \mathcal{M}$, $\gamma(0) = x$, $\gamma(1) = y$, $\gamma'(0) = v$

Convex optimization objective

$$\min f(x)$$
$$\text{subject to } x \in \mathcal{X} \subset \mathcal{M}$$

where $f : \mathcal{M} \to \mathbb{R} \cup \{\infty\}$, f is g-convex, $\mathcal{X}$ is a *geodesically convex set*, and $\mathcal{M}$ is a *Hadamard manifold*

- *Hadamard manifold*: Riemannian manifold that is simply connected with non-positive curvature
- **Cartan-Hadamard theorem**: if $\mathcal{M}$ is an n-dimensional Hadamard manifold, then it is diffeomorphic (both function and its inverse are continuously differentiable) to Euclidean space $\mathbb{R}^n$
- f assumes, at least, global minimum point within $\mathcal{X}$, $x^* \in \mathcal{X}$ is minimizer of f

Difficulties in non-asymptomatic analysis of Non-Euclidean spaces

- Consider: Euclidean update for optimization algorithms:
  $x_{s+1} = x_s - \eta_s g_s$,
  $\|x_{s+1} - x\|^2 = \|x_s - x\|^2 - 2\eta_s \langle g_s, x_s - x \rangle + \eta_s^2 \|g_s\|^2$; this does not hold for nonlinear spaces
- Consider: *linearization* of objective convex function f,
  $\psi(x; x_s) = f(x_s) + \langle g_s, x - x_s \rangle$, with unique solution $x_{s+1} = x_s - \eta_s g_s$ for $\min_x \{\psi(x; x_s) + \frac{1}{2\eta_s} \|x - x_s\|^2\}$
- **Recursive bound (Tseng 2009)**:
  $\psi(x_{s+1}; x_s) + \frac{1}{2\eta_s} \|x_{s+1} - x\|^2 \leq \psi(x; x_s) + \frac{1}{2\eta_s} \|x_s - x\|^2 - \frac{\eta_s}{2} \|g_s\|^2$
- No trivial analogy of linear function in nonlinear space, e.g. consider $\psi(x; y) = f(y) + \langle g_y, \mathrm{Exp}_y^{-1}(x) \rangle$ where $y \in \mathcal{M}$, $g_y \in T_y\mathcal{M}$, geodesically star-convex and star-concave in y, not convex nor concave in general

Adjustments to non-Euclidean properties

- Assumption: projection oracle $\mathcal{P}_\mathcal{X}$, $x \in \mathcal{M}$ mapped to $\mathcal{P}_\mathcal{X}(x) \in \mathcal{X} \subset \mathcal{M}$ s.t. $d(x, \mathcal{P}_\mathcal{X}(x)) < d(x, y)$ for all $y \in \mathcal{X} \setminus \{\mathcal{P}_\mathcal{X}(x)\}$
- *hyperbolic law of cosines*: $\cosh a = \cosh b \cosh c - \sinh b \sinh c \cos(A)$
- Trigonometric distance bounds
- **Lemma**: a, b, c sides of geodesic triangle in Alexandrov space (length space with curvature bound κ), A angle between b and c, then $a^2 \leq \xi(\kappa, c) b^2 + c^2 - 2bc \cos(A)$, where curvature-dependent property $\xi(\kappa, c) = \frac{\sqrt{|\kappa|} c}{\tanh(\sqrt{|\kappa|} c)}$
- Metric projection in Hadamard manifold is nonexpansive ($\|\mathcal{P}_\mathcal{X}(x) - \mathcal{P}_\mathcal{X}(y)\| \leq \|x - y\|$)

Lemma 7 (Bacák (2014)) *Let $(\mathcal{M}, \mathfrak{g})$ be a Hadamard manifold. Let $\mathcal{X} \subset \mathcal{X}$ be a closed convex set. Then the mapping $\mathcal{P}_\mathcal{X}(x) := \{y \in \mathcal{X} : d(x, y) = \inf_{z \in \mathcal{X}} d(x, z)\}$ is single-valued and nonexpansive, that is, we have for every $x, y \in \mathcal{M}$*

$$d(\mathcal{P}_\mathcal{X}(x), \mathcal{P}_\mathcal{X}(y)) \leq d(x, y).$$

Corollary 8 *For any Riemannian manifold $\mathcal{M}$ where the sectional curvature is lower bounded by κ and any point $x, x_s \in \mathcal{X}$, the update $x_{s+1} = \mathcal{P}_\mathcal{X}(\mathrm{Exp}_{x_s}(-\eta_s g_s))$ satisfies*

$$\langle -g_s, \mathrm{Exp}_{x_s}^{-1}(x) \rangle \leq \frac{1}{2\eta_s} \left(d^2(x_s, x) - d^2(x_{s+1}, x) \right) + \frac{\zeta(\kappa, d(x_s, x)) \eta_s}{2} \|g_s\|^2. \qquad (7)$$

Convergence rates

f	Algorithm	Stepsize	Rate[2]	Averaging[3]	Theorem
g-convex, Lipschitz	projected subgradient	$\frac{D}{L_f\sqrt{ct}}$	$O\left(\sqrt{\frac{c}{t}}\right)$	Yes	9
g-convex, bounded subgradient	projected stochastic subgradient	$\frac{D}{G\sqrt{ct}}$	$O\left(\sqrt{\frac{c}{t}}\right)$	Yes	10
g-strongly convex, Lipschitz	projected subgradient	$\frac{2}{\mu(s+1)}$	$O\left(\frac{c}{t}\right)$	Yes	11
g-strongly convex, bounded subgradient	projected stochastic subgradient	$\frac{2}{\mu(s+1)}$	$O\left(\frac{c}{t}\right)$	Yes	12
g-convex, smooth	projected gradient	$\frac{1}{L_g}$	$O\left(\frac{c}{c+t}\right)$	No	13
g-convex, smooth bounded variance	projected stochastic gradient	$\frac{1}{L_g+\frac{\sigma}{D}\sqrt{ct}}$	$O\left(\frac{c+\sqrt{ct}}{c+t}\right)$	Yes	14
g-strongly convex, smooth	projected gradient	$\frac{1}{L_g}$	$O\left(\left(1-\min\left\{\frac{1}{c},\frac{\mu}{L_g}\right\}\right)^t\right)$	No	15

Convergence Analysis: Nonsmooth convex optimization

Deterministic and stochastic subgradient methods achieve curvature-dependent $O(1/\sqrt{t})$ rate of convergence for g-convex on Hadamard manifolds.

Theorem: f g-convex and L_f-Lipschitz, diameter of domain bounded by D, sectional curvature $\kappa \leq 0$. Then, projected subgradient method, constant stepsize $\eta_s = \eta = \frac{D}{L_f \sqrt{\xi(\kappa, D) t}}$, $\bar{x}_1 = x_1$,

$\bar{x}_{s+1} = \mathrm{Exp}_{\bar{x}_s}(\frac{1}{s+1} \mathrm{Exp}_{\bar{x}_s}(x_{s+1}))$ satisfies $f(\bar{x}_t) - f(x^*) \leq DL_f \sqrt{\frac{\xi(\kappa, D)}{t}}$.

Theorem: f g-convex, diameter of domain bounded by D, sectional curvature $\kappa \leq 0$, stochastic subgradient oracle satisfies $\mathbb{E}[\tilde{g}(x)] = g(x) \in \partial f(x)$, $\mathbb{E}[\|\tilde{g}_s\|^2] \leq G^2$, then, projected stochastic subgradient method, stepsize $\eta_s = \eta = \frac{D}{G\sqrt{\xi(\kappa, D) t}}$, $\bar{x}_1 = x_1$,

$\bar{x}_{s+1} = \mathrm{Exp}_{\bar{x}_s}(\frac{1}{s+1} \mathrm{Exp}_{\bar{x}_s}(x_{s+1}))$ satisfies $\mathbb{E}[f(\bar{x}_t) - f(x^*)] \leq DG\sqrt{\frac{\xi(\kappa, D)}{t}}$.

Convergence Analysis: Strongly convex nonsmooth functions

Strongly convex nonsmooth functions. The following two theorems show that both subgradient method and stochastic subgradient method achieve a curvature dependent $O(1/t)$ rate of convergence for g-strongly convex functions on Hadamard manifolds.

Theorem 11 *If f is geodesically μ-strongly convex and L_f-Lipschitz, and the sectional curvature of the manifold is lower bounded by $\kappa \leq 0$, then the projected subgradient method with $\eta_s = \frac{2}{\mu(s+1)}$ satisfies*

$$f(\overline{x}_t) - f(x^*) \leq \frac{2\zeta(\kappa, D) L_f^2}{\mu(t+1)},$$

where $\overline{x}_1 = x_1$, and $\overline{x}_{s+1} = \mathrm{Exp}_{\overline{x}_s}\left(\frac{2}{s+1}\mathrm{Exp}_{\overline{x}_s}^{-1}(x_{s+1})\right)$.

Theorem 12 *If f is geodesically μ-strongly convex, the sectional curvature of the manifold is lower bounded by $\kappa \leq 0$, and the stochastic subgradient oracle satisfies $\mathbb{E}[\widetilde{g}(x)] = g(x) \in \partial f(x), \mathbb{E}[\|\widetilde{g}_s\|^2] \leq G^2$, then the projected subgradient method with $\eta_s = \frac{2}{\mu(s+1)}$ satisfies*

$$\mathbb{E}[f(\overline{x}_t) - f(x^*)] \leq \frac{2\zeta(\kappa, D) G^2}{\mu(t+1)}$$

where $\overline{x}_1 = x_1$, and $\overline{x}_{s+1} = \mathrm{Exp}_{\overline{x}_s}\left(\frac{2}{s+1}\mathrm{Exp}_{\overline{x}_s}^{-1}(x_{s+1})\right)$.

Convergence Analysis: Smooth convex optimization

Smooth convex optimization. The following two theorems show that gradient descent algorithm achieves a curvature dependent $O(1/t)$ rate of convergence, whereas stochastic gradient achieves a curvature dependent $O(1/t + \sqrt{1/t})$ rate for smooth g-convex functions on Hadamard manifolds.

Theorem 13 *If $f : \mathcal{M} \to \mathbb{R}$ is g-convex with an L_g-Lipschitz gradient, the diameter of domain is bounded by D, and the sectional curvature of the manifold is bounded below by κ, then projected gradient descent with $\eta_s = \eta = \frac{1}{L_g}$ satisfies for $t > 1$ the upper bound*

$$f(x_t) - f(x^*) \leq \frac{\zeta(\kappa, D) L_g D^2}{2(\zeta(\kappa, D) + t - 2)}.$$

Theorem 14 *If $f : \mathcal{M} \to \mathbb{R}$ is g-convex with L_g-Lipschitz gradient, the diameter of domain is bounded by D, the sectional curvature of the manifold is bounded below by κ, and the stochastic gradient oracle satisfies $\mathbb{E}[\widetilde{g}(x)] = g(x) = \nabla f(x), \mathbb{E}[\|\nabla f(x) - \widetilde{g}_s\|^2] \leq \sigma^2$, then the projected stochastic gradient algorithm with $\eta_s = \eta = \frac{1}{L_g + 1/\alpha}$ where $\alpha = \frac{D}{\sigma}\sqrt{\frac{1}{\zeta(\kappa, D) t}}$ satisfies for $t > 1$*

$$\mathbb{E}[f(\overline{x}_t) - f(x^*)] \leq \frac{\zeta(\kappa, D) L_g D^2 + 2D\sigma\sqrt{\zeta(\kappa, D) t}}{2(\zeta(\kappa, D) + t - 2)},$$

where $\overline{x}_2 = x_2$, $\overline{x}_{s+1} = \mathrm{Exp}_{\overline{x}_s}\left(\frac{1}{s}\mathrm{Exp}_{\overline{x}_s}^{-1}(x_{s+1})\right)$ for $2 \leq s \leq t-2$, $\overline{x}_t = \mathrm{Exp}_{\overline{x}_{t-1}}\left(\frac{\zeta(\kappa, D)}{\zeta(\kappa, D) + t - 2}\mathrm{Exp}_{\overline{x}_{t-1}}^{-1}(x_t)\right)$.

Convergence Analysis: Smooth and strongly convex functions

Smooth and strongly convex functions. Next we prove that gradient descent achieves a curvature dependent linear rate of convergence for geodesically strongly convex and smooth functions on Hadamard manifolds.

Theorem 15 *If $f : \mathcal{M} \to \mathbb{R}$ is geodesically μ-strongly convex with L_g-Lipschitz gradient, and the sectional curvature of the manifold is bounded below by κ, then the projected gradient descent algorithm with $\eta_s = \eta = \frac{1}{L_g}$, $\epsilon = \min\{\frac{1}{\zeta(\kappa,D)}, \frac{\mu}{L_g}\}$ satisfies for $t > 1$*

$$f(x_t) - f(x^*) \leq \frac{(1-\epsilon)^{t-2} L_g D^2}{2}.$$

Matrix Karcher mean problem

For set of N symmetric positive definite matrices $\{A_i\}_{i=1}^N$, Karher mean is defined as PSD matrix minimizing sum of squared distance induced by Riemannian metric, $d(X, Y) = \|\log(X^{-1/2}YX^{-1/2})\|_F$, with loss function $f(X; \{A_i\}_{i=1}^N) = \sum_{i=1}^N \|\log(X^{-1/2}A_iX^{-1/2})\|_F^2$ nonconvex in Euclidean space, geometrically $2N$-strongly convex. Gradient update step,

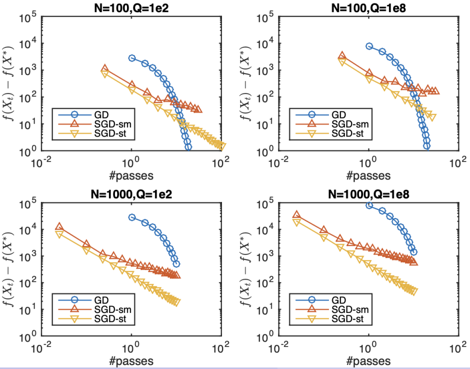
$$X_{s+1} = X_s^{1/2} \exp\left(-\eta_s \sum_{i=1}^N \log(X_s^{1/2}A_i^{-1}X_s^{1/2})\right) X_s^{1/2}$$

SGD, with random $i(s)$ from $\{1, ..., N\}$,

$$X_{s+1} = X_s^{1/2} \exp\left(-\eta_s N \log(X_s^{1/2}A_{i(s)}^{-1}X_s^{1/2})\right) X_s^{1/2}$$

Comparison of performance

Zhang and Sra compared three algorithms: gradient descent (GD) with $\eta_s = \frac{1}{5N}$; stochastic gradient method for smooth functions (SGD-sm), $\eta_s = \frac{1}{L_g + 1/\alpha}$, $\alpha = \frac{D}{\sigma}\sqrt{\frac{1}{\xi(\kappa, D)t}}$; Stochastic gradient method for strongly convex functions (SGD-st), $\eta_s = \frac{1}{N(s+1)}$ set w.r.t. $2N$-strong convexity of loss function.



Discussion

- Empirical observation that L_g estimate of $5N$ guarantees convergence
- GD converages at linear rate, SGD-sm asymtotically at $O(1/\sqrt{t})$ rate, SGD-st at $O(1/t)$ rate

References

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